

Agenda

- Lab 2 reviews
- Intro to Virtual Memory
 - Bit Manipulation / Math Refresher
 - Page Tables
 - Multi-Level Page Tables
 - Alternatives + Trade offs
 - Introduce VM in x86-64 arch.

Last Week:

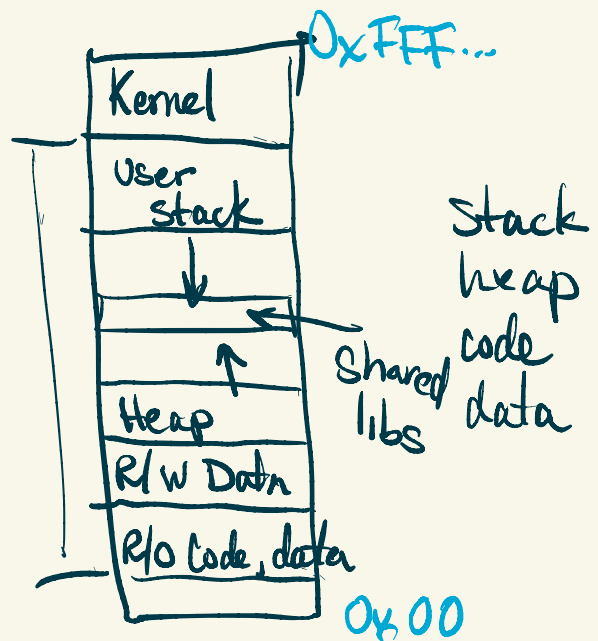
- Deadlocks
- Different ways to prevent deadlock
- Preconditions for deadlock.
- Monitors

OSTEP

- Virtualization
- Concurrency
- Persistence
- Security

Virtual Memory

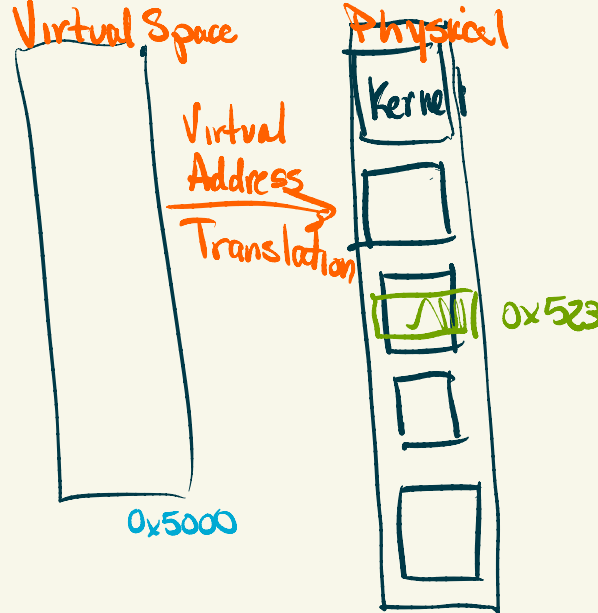
Virtual Addresses →
Physical Addresses



VA $\rightarrow$ PA
Virtual Address Physical Address

Why / Benefits:

- Programmability
- Protection
- Effective Use of Resources
- Sharing



Bit Math Refresher

Hexadecimal (Base 16)

0x 0 1 2 3 4 5 6 7 8 9 A B C D E F

0x0 $\Rightarrow$ 0b 0000

0b 1111 $\Rightarrow$ 0xf

2^3 2^2 2^1 $2^0 = 15$

32-bit CPU

How many addresses?

$\sim 4GB \Rightarrow 2^{10} = 1k \quad 4G$

64-bit CPU addresses = $2^{64} \quad 2^4 \cdot 2^{60} = 16 \text{ Ex}$

0x 16

$2^0 = 1$

$2^1 = 2$

$2^2 = 4$

$2^3 = 8$

$2^4 = 16$

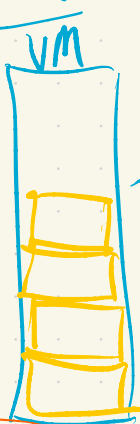
$2^5 = 32$

$2^6 = 64$

$2^7 = 128$

PAGING

Page 4KB.
 $\Rightarrow 2^{12}$



PM

Pages Have #s
 Page Number

4095
 0

PN = 1

PN = 0

$2^{10} = 1K$ kilo
 $2^{20} = 1M$ mega
 $2^{30} = 1G$ giga
 $2^{40} = 1T$ tera
 $2^{50} = 1P$ peta
 $2^{60} = 1E$ exa

32-bit CPU

2^{32}

4KB pagesize 2^{12}

How many pages? 2^{20}

$$2^{32} / 2^{12} = 2^{20}$$

48 bits for addresses? $2^{48} / 2^{12} = 2^{36}$
 $\Rightarrow 64B.$

VPN: Virtual Page Number

PPN: Physical Page Number

of Possible Virtual Addresses

n: # address bits N: # Virtual add. Largest Possible Virtual Address

8	$2^8 = 256$	255
16	$2^{16} = 64K = 2^{10} \cdot 2^{6}$	$64K - 1$
32	2^{32}	$2^{32} - 1 = 4G - 1$
	$2^? = 256T$	
64		

$$\begin{aligned}
 2^{32} &= \\
 2^{30} \cdot 2^2 &= \\
 4 \cdot 2^{30} &=
 \end{aligned}$$

Page Table

- An index!

Virtual Address

VPN	PPN
0x000	0x538



VPN

offset

Page Size: $4K = 2^{12}$

VA: 32 bits.

Page Size 4k

32-bit addresses.

0b1100100011... (32)

VA:

0x123456

PA:

0xab9876

NO!

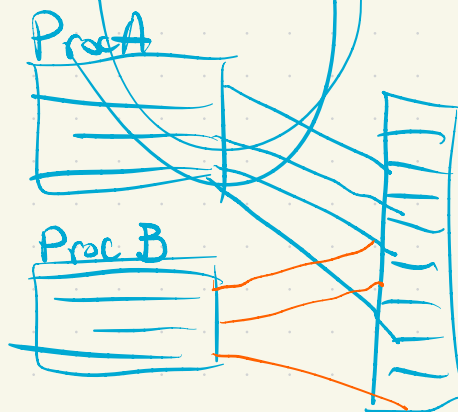
0. $\Rightarrow$ VPN=0

PPN = 0
PPN = 1
PPN = 2
...

$\Downarrow$ offset of 4 bits
 $\Rightarrow$ page size is $2^4 = 16$

2^{20} entries.

Assume: 8 bytes/entry



Example:

48-bit addresses

4 KB page size

- How many PTEs?

$$2^{36}$$

- 8 bytes / PTE

$$2^{36} \cdot 2^3 = 2^{39} \rightarrow 2^{30} \cdot 2^9 = 512 \text{ GB}$$

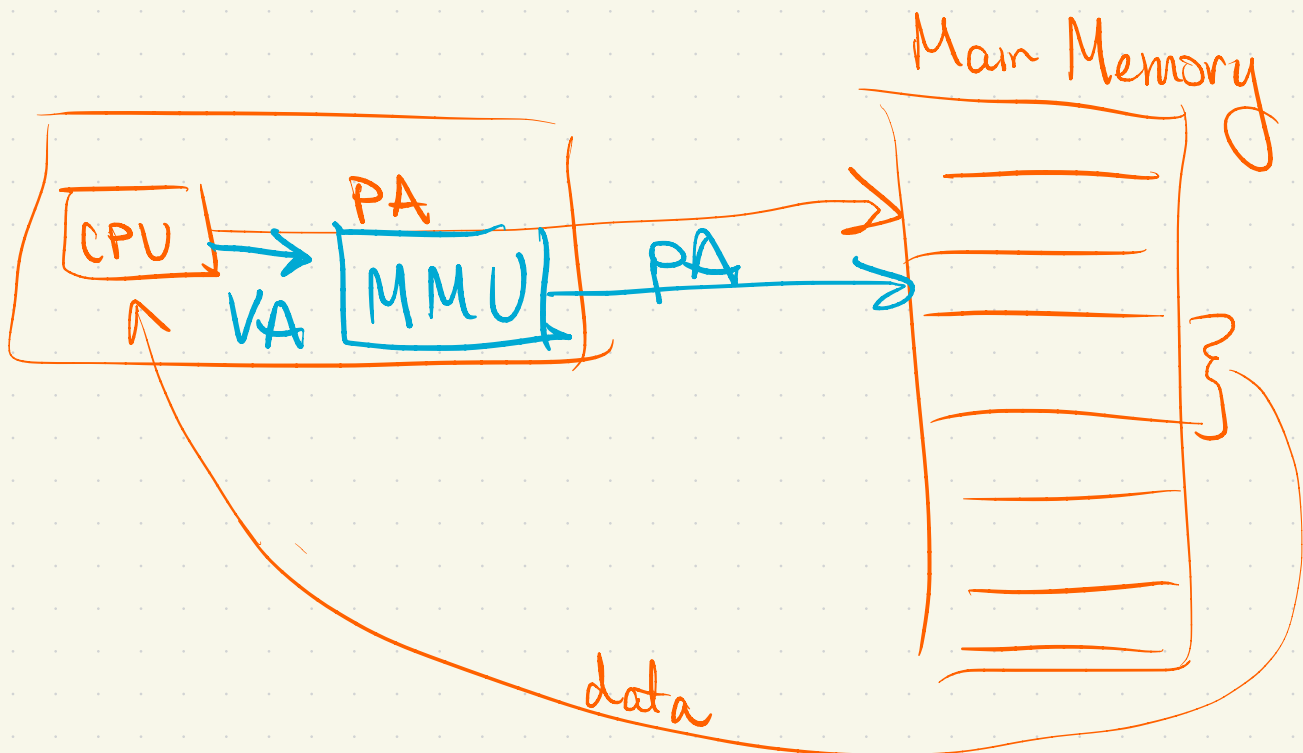
▶ What is in the PTE?

- PPN $\Rightarrow$ Physical Page Num

36 bits $\Rightarrow$ 4.5 bytes

- "dirty" bit

- "invalid"



(Bits in VA)

n

16 $\Rightarrow 2^{16}$ addresses

Page Size

$$P = 2^p$$

$$4K = 2^{12}$$

PTEs

$$16 \equiv 2^{16} / 2^{12} =$$

16 $\Rightarrow 2^{16}$ addresses

8K $\Rightarrow 2^{13}$ size

$$2^{16} / 2^{13} = 2^3 = 8$$

2⁴ = 16.
PTEs.

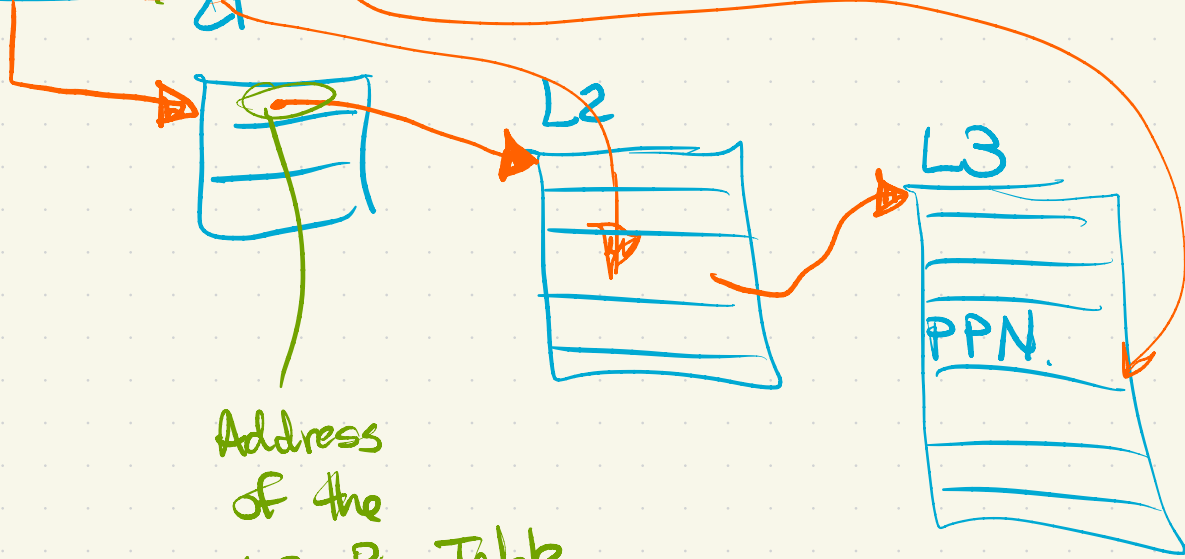
32-bit virtual address space, 24 bit physical address space.

page size	<u>P</u>	<u>VPN bits</u>	<u>VPO bits</u>	<u>PPN bits</u>	<u>PPO bits</u>
1 KB = 2 ¹⁰		22	10	14	10

MultiLevel Page Table.

VA: VPN VPO

L1PN / L2PN / L3PN | VPO



Address
of the
L2 Page Table

$PPN + VPO$
↓
PA.

x86-64